



Using materials, components and equipment provided, perform the following tasks:

- Potentiometer

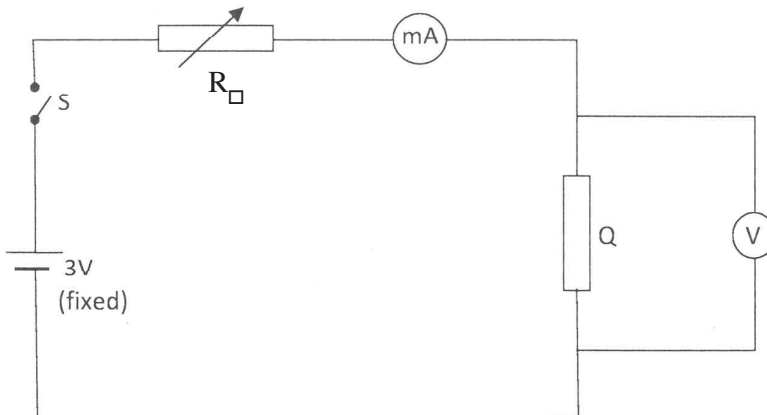


Figure 1

- ☐  Close Switch 
- ☐  Adjust the Potentiometer for the Ammeter to obtain current values in Table 11 and in each
- ☐  case record the corresponding voltage values    1 2 marks

Table 11

Current $\square_{\text{mA}} \square$	$\square \square$	$\square \square \square$	$\square \square \square$	$\square \square \square$	$\square \square \square$
Voltage $\square \square \square \square \square$					
$\frac{V}{I}$					

- ☐ **iv** Calculate the values of $\frac{1}{I}$ and record them in the spaces provided in the table
- ☐ **v** Use the values in the table to draw a graph of Voltage against Current 1/2 marks

(vi) Determine the slope of the graph. (2 marks)

- ☐ ☒ vii From The Graph Determine The Voltage V When The Current Is 0.004 A

☐ ☐ $V =$ _____ mark ☐

☐ ☒ viii State The Purpose Of The Experiment _____ mark ☐

- [illegible]



EXERCISE III

Use the Tools, Equipment and Materials provided to make the Bracket shown in Figure 2

10 Marks

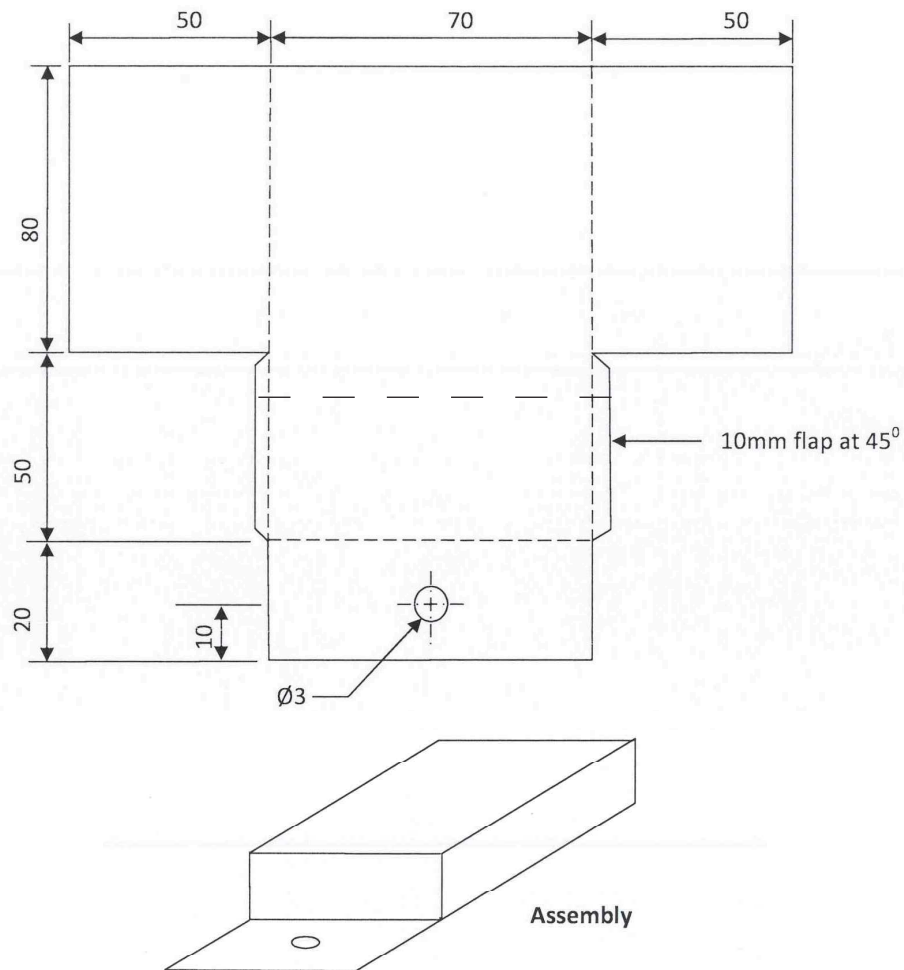


Figure 2

EXERCISE III

Figure 1 shows a block diagram of the electronic circuit provided

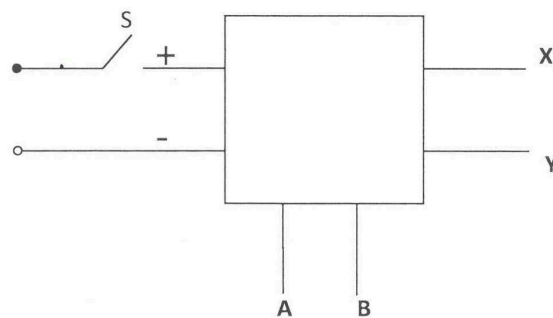


Figure 1



Perform the following tasks

(a) With the switch S open, connect the circuit to the DC power source. Let the examiner check your work 3 marks

(b) Set the potentiometer to $0\ \Omega$ as measured with ohmmeter. Select a value of resistor shown in Table 1 and in each case do the following

connect each resistor between terminals A and B 1/2 marks

close the switch, measure the voltage and record in Table 1 3 marks

Table 1

Resistor (Ω)	V_R	$P = \frac{V_R^2}{R}$
10		
20		
30		
40		
50		

Calculate the power dissipated by each resistor and complete Table 1 3 marks

(iv) Draw a graph of power against resistance. 5 1/2 marks

From the graph determine the value of R for which there is maximum power transfer 3 marks

EXERCISE

Use the components and equipment to connect the circuit illustrated in Figure 4. Let the examiner see your work 1/2 marks

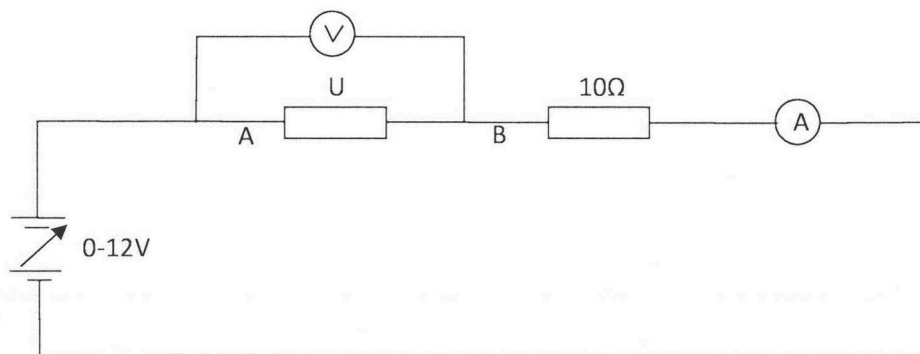


Figure 4



☐ Perform the following tasks

☐ a. Switch on the power supply

☐ b. Adjust the power supply to obtain each of the voltage values across 10Ω as shown in

☐ ☐ table and in each case record the corresponding current ☐ ☐ marks

Table

V					
I					

☐ c. Switch off the power supply and reverse its connections

☐ d. Adjust the power supply to obtain each of the voltage values across 10Ω as shown in

☐ ☐ table and in each case record the corresponding current ☐ ☐ marks

Table

V					
I					

☐ e. Use the values of I and V from Tables and to draw the graph of current against

☐ ☐ voltage 10Ω in the same axes ☐ ☐ marks

☐ f. From the shape of the graph identify component ☐ ☐

☐ marks

EXERCISE

☐ ☐ Figure shows the layout of a lighting installation using PVC sheathed cables. Install the

☐ circuit such that the lamps are controlled at one point ☐ ☐ ☐ ☐

☐ marks

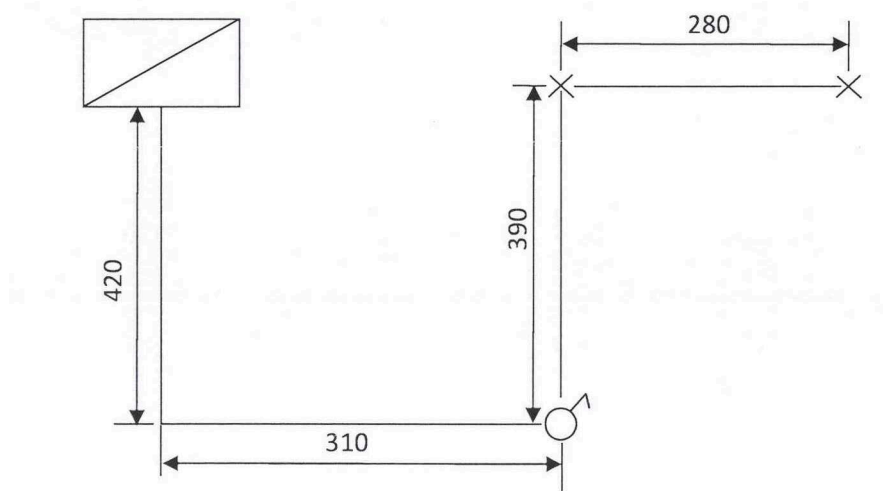


Figure 5